



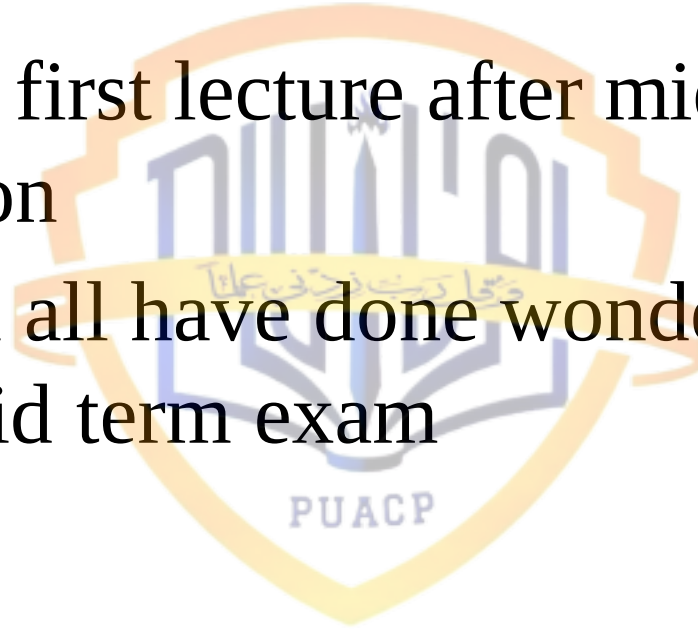
Software Testing – 1

- Introduction

Lecture # 23

Recap

- This is the first lecture after mid term examination
- I hope you all have done wonderfully well on your mid term exam



Today's Lecture

- Starting today, we'll be starting a series of lectures on software testing and the role of software testing in software quality assurance and quality control
- So, let's begin

Basics of Software Testing - 1

- Synonyms: checking, assessing, examination
- Testing is one of the most important parts of quality assurance and the most performed QA activity

Basics of Software Testing - 2

- Testing has been a basic form of defect removal since software industry began
- For majority of software projects, it is the *only* form of defect removal utilized

Basics of Software Testing - 3

- The basic idea of testing involves the execution of software and the observation of its behavior or outcome
- If a failure is observed, the execution record is analyzed to locate and fix the faults that caused the failure
- Otherwise, we gain some confidence that the software under testing is more likely to fulfill its designated functions

Why Software Testing? - 1

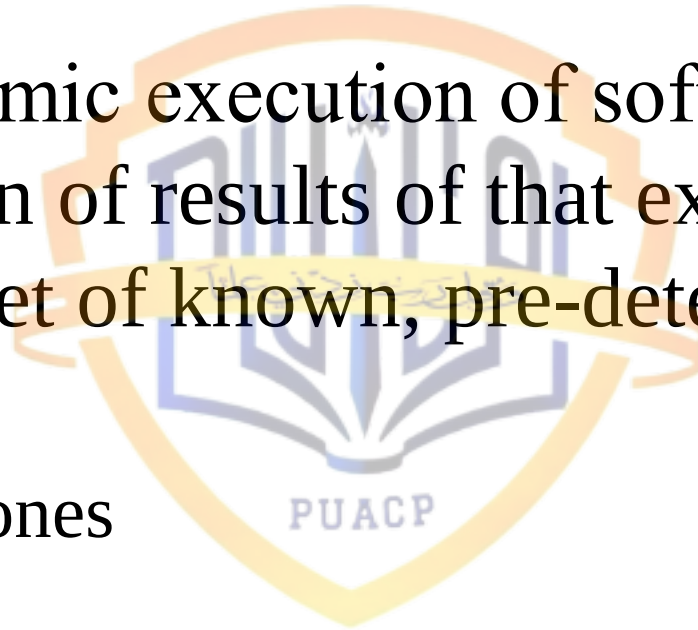
- The purpose of software testing is to ensure that the software systems would work as expected when they are used by their target customers and users
- For software products, software testing provides a mechanism to demonstrate their operation through controlled execution

Software Testing

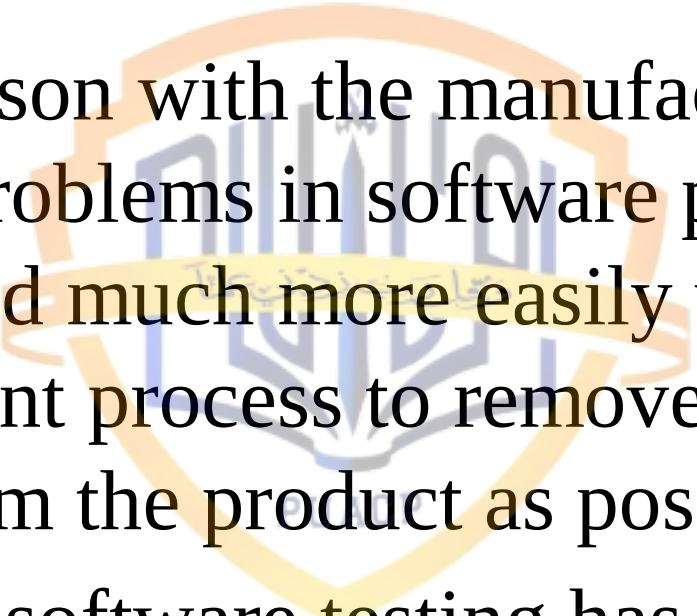


Software Testing

- “The dynamic execution of software and the comparison of results of that execution against a set of known, pre-determined criteria”
 - Capers Jones



- This demonstration of proper behavior is a primary purpose of software testing, which can also be interpreted as providing evidence of quality in the context of software quality assurance, or as meeting certain quality goals

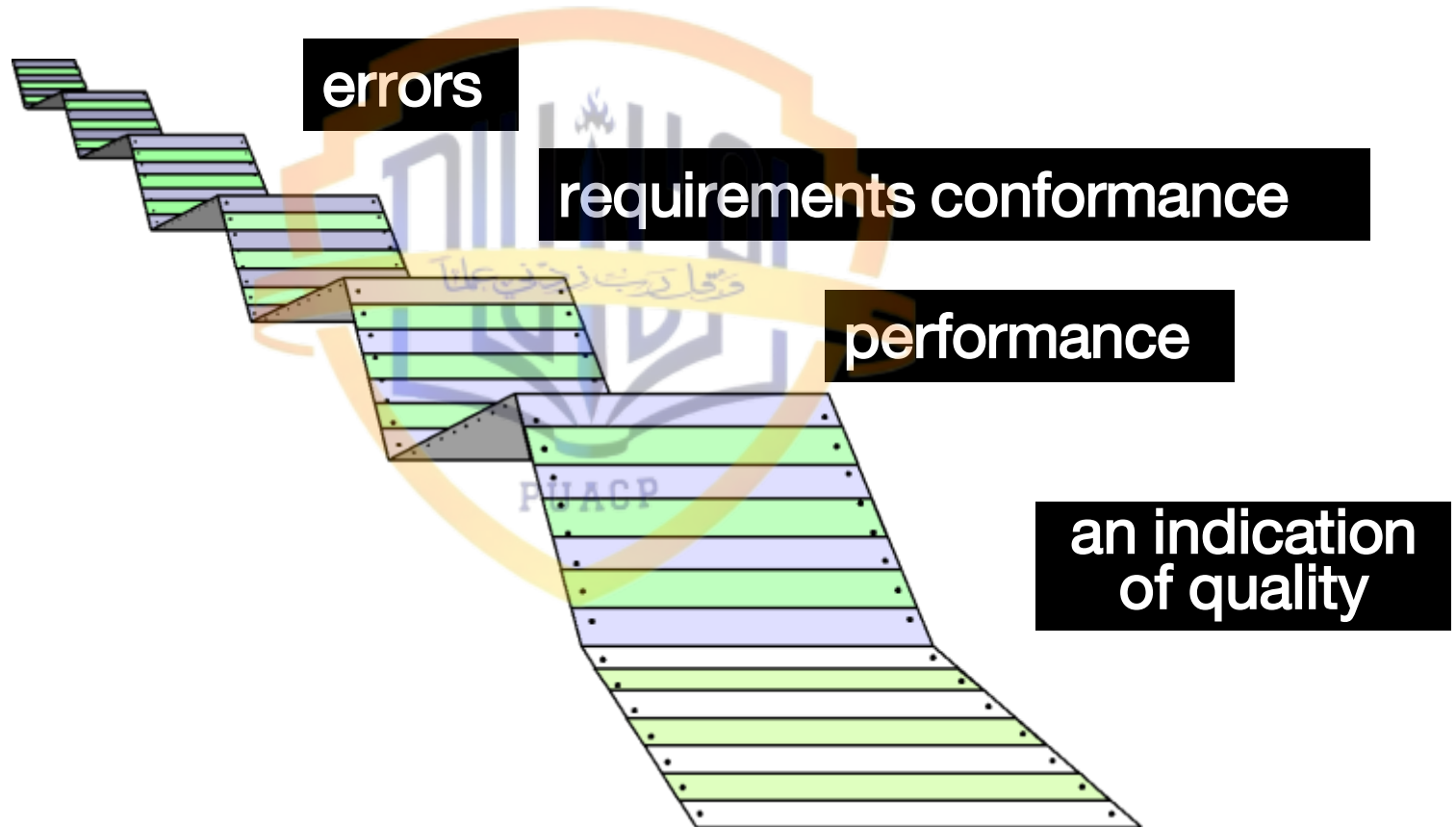
- 
- In comparison with the manufacturing systems, problems in software products can be corrected much more easily within the development process to remove as many defects from the product as possible
 - Therefore, software testing has two primary purposes

Primary Purposes of Software Testing

- To demonstrate quality or proper behavior
- To detect and fix problems

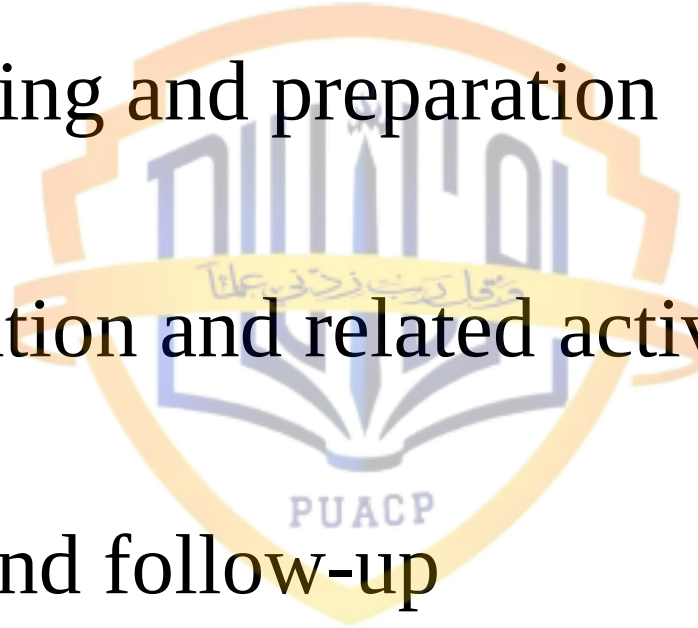


What Testing Shows?



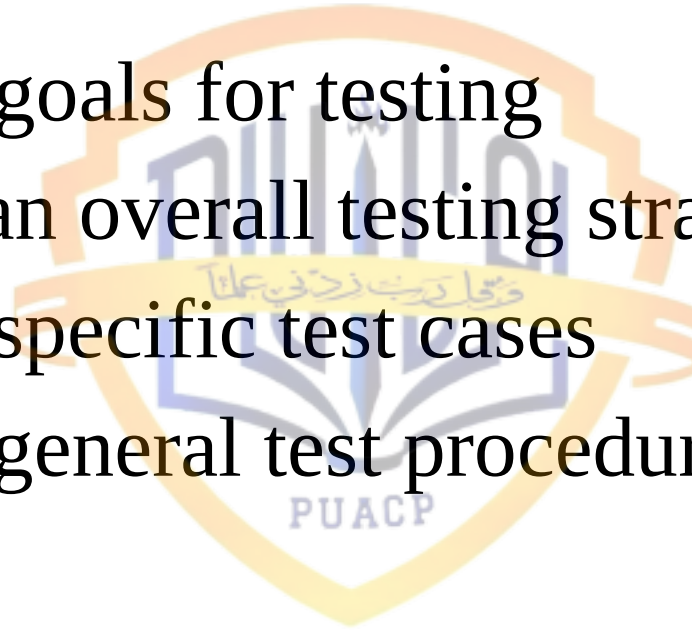
Major Activities and Generic Testing Process

- Test planning and preparation
- Test execution and related activities
- Analysis and follow-up



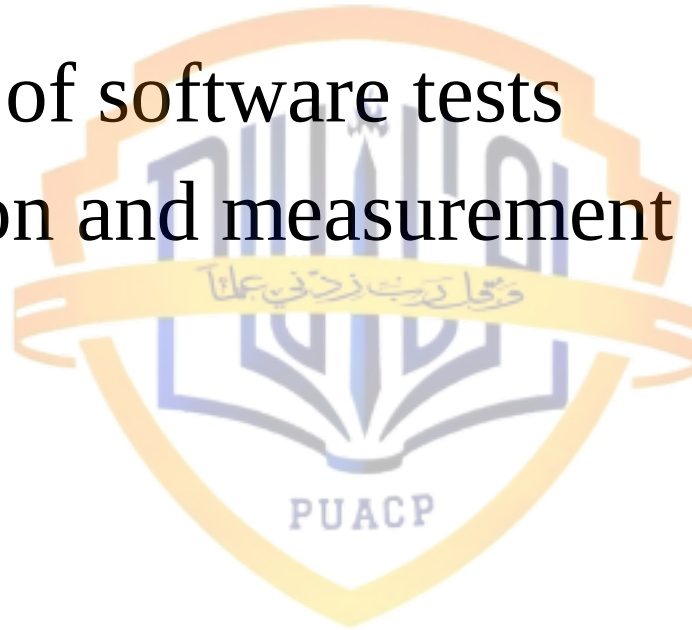
Test Planning and Preparation

- Setting of goals for testing
- Selecting an overall testing strategy
- Preparing specific test cases
- Preparing general test procedure



Test Execution

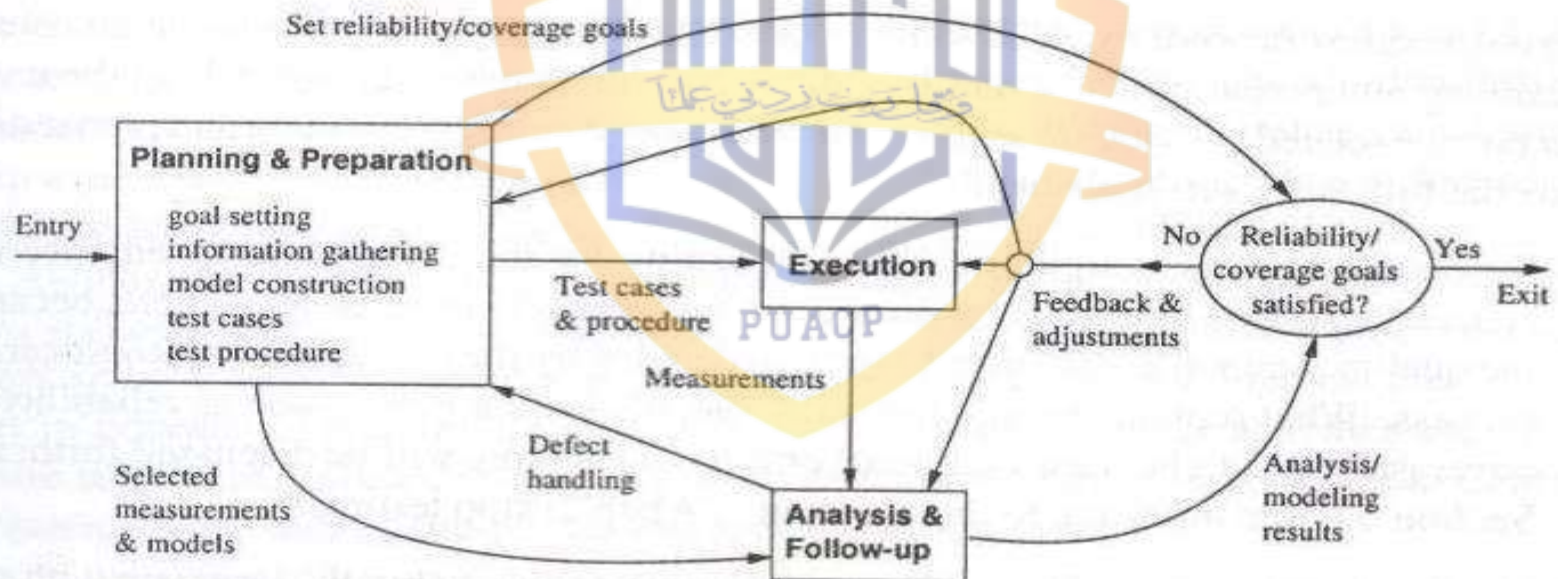
- Execution of software tests
- Observation and measurement of product behavior

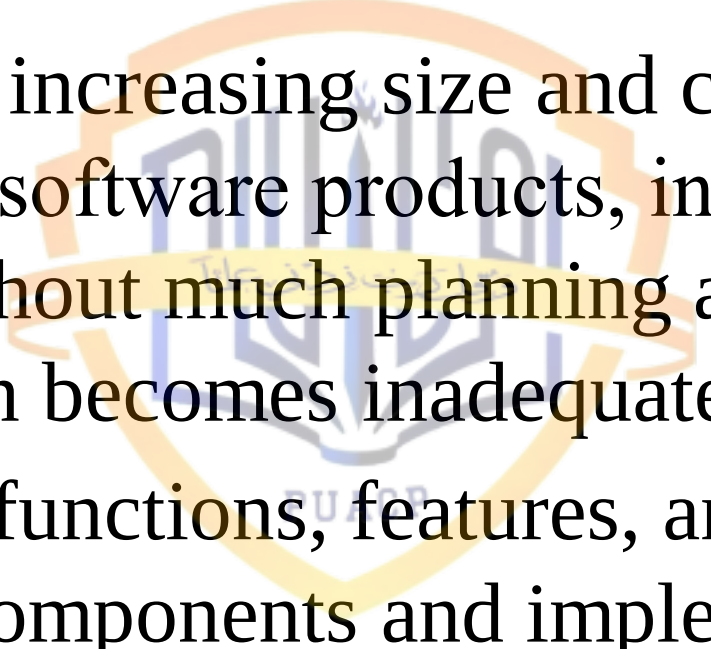


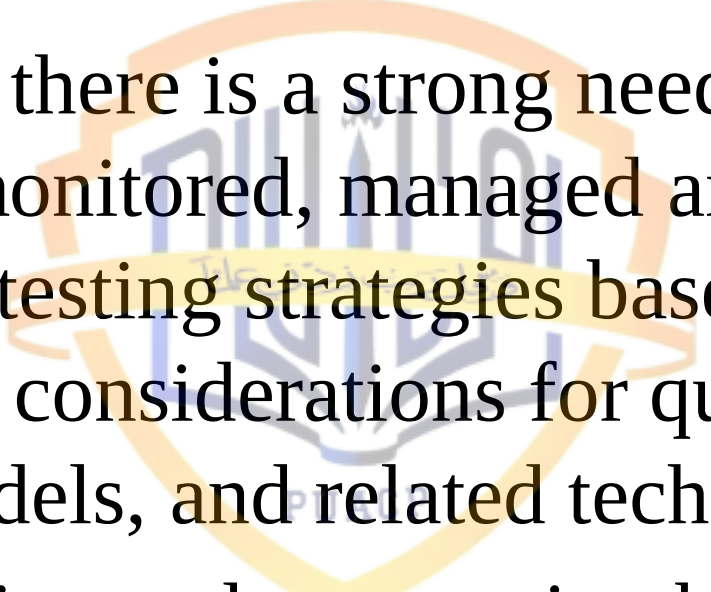
Analysis and Follow-Up

- Includes result checking and analysis to determine if a failure has been observed, and if so, follow-up activities are initiated and monitored to ensure removal of the underlying causes, of faults, that led to the observed failures in the first place

Generic Testing Process



- 
- Due to the increasing size and complexity of today's software products, informal testing without much planning and preparation becomes inadequate
 - Important functions, features, and related software components and implementation details could be easily overlook in such informal testing

- 
- Therefore, there is a strong need for planned, monitored, managed and optimized testing strategies based on systematic considerations for quality, formal models, and related techniques
 - Test planning and preparation has these sub-activities

Sub-Activities in Test Planning and Preparation

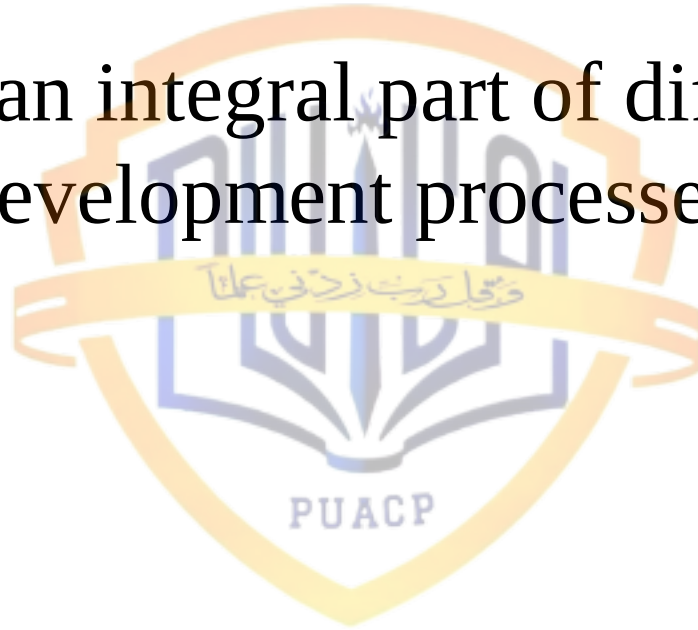
- Goal setting
 - Reliability and coverage goals
- Test case preparation
 - Constructing new test cases or generating automatically, selecting them from existing ones for legacy products, and organizing them in some systematic manner
- Test procedure preparation
 - It is defined and followed to ensure effective test execution, problem handling and resolution, and overall test process management

Testing as a Part of QA in Overall Software Process - 1

- Testing and inspections often find different kinds of problems, and may be more effective under different circumstances
- Therefore, inspections and testing should be viewed more as complementary QA alternatives instead of competing ones

Testing as a Part of QA in Overall Software Process - 2

- Testing is an integral part of different software development processes



Testing and Defect Removal Efficiency

- Most forms of testing are less than 30% efficient in finding bugs
- Many forms of testing should be used in combination with pre-test design and code inspections to achieve high defect removal efficiencies

Testing and Defect Types

- Testing is much more effective in finding errors of commission, or things that are done wrong, than it is in finding errors of omission or things that are left out by accident



Questions about Testing



Basic, testing techniques, testing activities and management

Basic Questions about Testing

- Our basic questions about testing are related to the objects being tested, perspectives and views used in testing, and overall management and organization of test activities

Basic Questions about Testing

- What artifacts are tested?
- What to test, and what kind of faults is found?
- When, or at what defect level, to stop testing?
 - Coverage information, product reliability goals,

Questions about Testing Techniques

- Many different testing techniques can be applied to perform testing in different sub-phases, for different types of products, and under different environments
- Various questions regarding these testing techniques can help us get a better understanding of many related issues

Questions about Testing Techniques - 1

- What is the specific testing technique used?
- What is the underlying model used in a specific testing technique?
 - There are two basic models: 1) based on simple structures such as checklists and partitions, and 2) based on finite-state machines

Questions about Testing Techniques - 2

- Are techniques for testing in other domains applicable to software testing?
 - This defect seeding, mutation, immunization used in physical domains and comes under specialized testing in software
- If multiple testing techniques are available, can they be combined or integrated for better effectiveness or efficiency?

Questions about Test Activities and Management

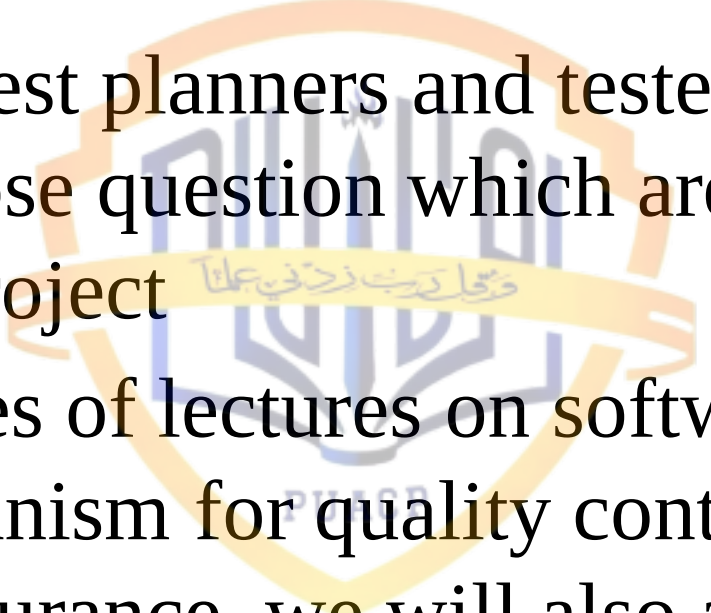
- Various other questions can also be used to help us analyze and classify different test activities
- Some key questions are about initiators and participants of these activities, organization and management of specific activities
- (Eight questions)

Questions about Test Activities and Management - 1

- Who performs specific activities?
- When can specific test activities be performed?
- What process is followed for these test activities?
- Is test automation possible? And if so what kind of automated testing tools are available and usable for specific applications?

Questions about Test Activities and Management - 2

- What artifacts are used to manage the testing process and related activities?
- What is the relationship between testing and various defect-related concepts?
- What is the general hardware /software / organizational environment for testing?
- What is the product type or market segment for product under testing?

- 
- Software test planners and testers need to answer those question which are relevant for their project
 - In the series of lectures on software testing, as a mechanism for quality control and quality assurance, we will also address them

References

- Software Quality Engineering: Testing, Quality Assurance, and Quantifiable Improvement by Jeff Tian, IEEE Computer Society, 2005 (Chapter 6.1-6.2)